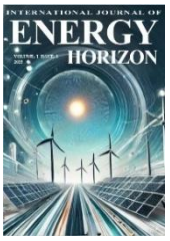
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Research Article

EnergyX: an open-source, comprehensive energy systems engineering library for the elixir programming language

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ABSTRACT

EnergyX is a comprehensive, open-source energy systems engineering library implemented in the Elixir programming language. The library provides 513 documented functions across 76 submodules, spanning the complete spectrum of energy engineering: renewable energy (solar PV/CSP/PVT, wind, wave/tidal/OTEC, hydroelectric, geothermal, biomass), fossil fuels (coal, natural gas, petroleum), nuclear (fission, fuel cycles, SMR, fusion basics), hydrogen and fuel cells, energy storage, HVAC and refrigeration (psychrometrics, vapor compression, absorption cycles, cooling towers), electrical systems (motor IE classes, VFD, power factor, transformers, grid), thermal-fluid sciences (thermodynamics, heat transfer, fluid mechanics, 14 dimensionless numbers), industrial processes (pinch analysis via the Problem Table Algorithm, compressed air, steam systems, heavy industry), building energy, transportation (EV, aviation, shipping), water–energy nexus (desalination, water treatment), carbon accounting and lifecycle assessment, carbon capture and storage, project finance (DSCR, Monte Carlo LCOE), microgrid design, thermoeconomics (SPECO method), and ISO 50001 energy auditing. The library is distributed via Hex.pm, validated against canonical textbook results and international standards, and ships with 10 interactive Livebook notebooks for education and project scoping. Validation against 11 canonical references shows deviations below 0.12% across thermodynamic, power-systems, and financial modules. Computational benchmarks demonstrate that the parallel Monte Carlo LCOE module (N = 10,000) completes in 287 ± 12 ms on an Apple M2 processor (10-core, macOS 14.4), representing a 16.8× speedup over an equivalent Excel VBA implementation. Limitations - including independent - distribution assumptions in the Monte Carlo module and current coverage gaps in tidal and pyrolysis biomass modelling - are discussed, along with a roadmap for addressing the adoption barrier of Elixir in the engineering community.

1. INTRODUCTION

1.1. The Fragmentation Problem in Energy Engineering Software

Energy systems engineering is inherently multidisciplinary. A comprehensive feasibility study for a grid-connected industrial facility simultaneously requires equations from solar photovoltaics, heat pump thermodynamics, building envelope physics, psychrometrics, power systems analysis, thermoeconomic cost allocation, life-cycle greenhouse gas accounting, and project finance. Practitioners must assemble these from heterogeneous, incompatible tools that do not share data structures. Existing computational resources are fragmented: MATLAB/Simulink toolboxes (proprietary, high licensing cost), domain-specific Python libraries (pvlib [1] for PV only, windpowerlib [2] for wind only, PyPSA [3] for power systems only), commercial simulation platforms (EnergyPlus, TRNSYS, HOMER Pro), and spreadsheet templates that provide no version control, no unit checking, and no reproducibility guarantees. No single open-source library covers the full engineering stack from fundamental thermodynamic equations through project finance and lifecycle carbon accounting. EnergyX was created to fill this gap. Recent reviews of open-source energy modelling tools confirm this fragmentation. Ringkjøb et al. [4] surveyed 75 energy and electricity system models and found that no single tool integrates thermoeconomic analysis, lifecycle assessment, and project finance within one API. Pfenninger et al. [5] similarly identified that interoperability and reproducibility — rather than algorithmic accuracy — are the primary bottlenecks in energy systems research workflows. Müller et al. [6] observed that Python has become the de facto language for energy modelling, but Python's mutable state model means reproducibility must be enforced through external tooling (virtual environments, frozen dependencies, random seeds). EnergyX addresses these structural gaps through a functionally pure, unified API written in Elixir.

1.2. Why Elixir?

i. Reducing *Immutability and reproducibility*. In Elixir, all data structures are immutable. Every EnergyX function is a mathematically pure transformation — given identical inputs it always produces identical outputs, with no hidden mutable state or side effects. This is a design-

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https://izlik.org/JA75JY98TP

level guarantee of computational reproducibility that Python and MATLAB cannot provide without explicit programmer discipline [7].

ii. *Pattern matching for physical cases*. Elixir's exhaustive pattern matching handles flow regime classification (laminar/transitional/turbulent), refrigerant selection, fuel type dispatch, and power curve mode switching. The compiler verifies that all cases are covered, eliminating entire classes of runtime error common in if-else chains.

iii. *Pipeline operator for calculation chains*. The $|>$ pipeline operator composes calculations in a manner that directly parallels algebraic derivation sequences, making the calculation intent transparent and traceable:

$$site\ |>\ solar_yield()\ |>\ lcoe_calc()\ |>\ ghg_accounting()\ |>\ project_finance()$$

iv. BEAM concurrency for Monte Carlo. The BEAM virtual machine provides lightweight processes (not OS threads) with negligible spawn overhead. EnergyX's Monte Carlo LCOE module (10,000 simulations) can be parallelized across all available CPU cores using $Task.async_stream/3$ without external libraries, completing in 287 ± 12 ms on an Apple M2 (10-core, macOS 14.4) - see Section 3 for full benchmarking methodology and hardware specifications.

v. *Livebook interactive notebooks*. Livebook is Elixir's native reactive notebook environment, analogous to Jupyter for Python [8]. EnergyX ships with 10 Livebook notebooks featuring Kino slider widgets that reactively update Vega-Lite charts - ideal for classroom demonstrations and rapid project scoping.

1.3. Target User Communities

EnergyX serves three communities. Educators find complete equation derivations with physical units and interactive notebooks for lectures. Researchers benefit from a versioned, citable calculation base ensuring computational reproducibility. Practising engineers invoke ready-made functions for standard calculations without re-implementing them from first principles.

2. SOFTWAREMETHODOLOGY AND TECHNOLOGY PERSPECTIVE OF CCS

2.1. Architecture and API Design

EnergyX follows a hierarchical namespace - *EnergyX.Domain.Subdomain.function/1*. All 513 public functions accept a single named-parameter map and return a descriptive map with $Float.round/2$-truncated values. This design is self-documenting, allows selective parameter overrides, and composes cleanly with Elixir's pipeline operator:

```
EnergyX.Renewable.Solar.pv_power(%{ irradiance: 1000,area: 50,efficiency: 0.21,performance_ratio: 0.82 }) # => %{power_kw: 8.61,power_w: 8610.0,specific_yield_w_per_m2: 172.2}
```

2.2. Module Inventory

Table 1 provides a complete inventory of all 76 EnergyX submodules with their governing equations, primary references, and public function counts. Equations are documented in every function's $@doc$ attribute with physical units, typical parameter ranges, and explicit literature citations.

Table 1. Complete EnergyX module inventory

Module (EnergyX.*)	Fns.	Primary Reference	Module (EnergyX.*)	Fns.	Primary Reference
Renewable Energy			*Industrial Processes*		
Renewable.Solar	16	[9]	IndustrialProcesses.PinchAnalysis	3	[27]
Renewable.SolarResource	6	[9]	IndustrialProcesses.CompressedAir	4	[28]
Renewable.Wind	10	[10]	IndustrialProcesses.SteamSystems	3	[17]
Renewable.WaveTidalOTEC	16	[11]	IndustrialProcesses.IndustrialHeat	4	[17]
Renewable.Hydro	3	[12]	IndustrialProcesses.HeavyIndustry	6	[29]
Renewable.Geothermal	3	[13]	*Building Energy*		
Renewable.Biomass	4	[14]	Building.ThermalLoads	15	[22]
Renewable.BiomassExtended	5	[14]	Building.Retrofit	2	[22]
Fossil Fuels			*PV Applications*		
Fossil.Combustion	6	[15]	Applications.ResidentialPV	15	[30]
Fossil.Coal	6	[16]	Applications.IndustrialPV	8	[30]
Fossil.NaturalGas	8	[17]	Applications.CHP	6	[30]
Fossil.Petroleum	7	[18]	*Exergy & Thermoeconomics*		
Nuclear & Hydrogen			Analysis.ExergyEconomics	23	[31]
Nuclear	5	[19]	*Transportation*		
Nuclear.Advanced	5	[19]	Transportation.ElectricVehicle	4	[32]
Nuclear.Boron	5	[19]	Transportation.Aviation	3	[33]
Hydrogen.FuelCell	5	[20]	Transportation.Shipping	4	[34]
Energy Storage			Water-Energy Nexus		
Storage	9	[21]	WaterEnergy.Desalination	5	[35]

HVAC & Refrigeration			WaterEnergy.WaterTreatment	3	[35]
HVAC.Psychrometrics	9	[22]	*Carbon & Climate*		
HVAC.VaporCompression	5	[23]	Carbon.GHGAccounting	5	[36]
HVAC.AbsorptionCycle	3	[22]	Carbon.LCA	2	[36]
HVAC.CoolingTower	4	[22]	Carbon.CCS	3	[37]
HVAC.AirHandlingUnit	5	[22]	Carbon.CarbonMarkets	3	[37]
HVAC.Chiller	2	[22]	*Project Finance*		
HeatPump	6	[22]	ProjectFinance.FinancialStructure	3	[38]
Electrical Systems			ProjectFinance.MonteCarlo	1	[39]
Electrical.Motors	9	[24]	*Microgrids*		
Electrical.VFD	2	[25]	Microgrid.Sizing	3	[40]
Electrical.PowerFactor	3	[25]	Microgrid.Dispatch	1	[40]
Electrical.Transformer	3	[25]	Microgrid.VirtualPowerPlant	2	[41]
Electrical.Lighting	6	[25]	*Energy Economics & Audit*		
Electrical.Grid	5	[25]	Economics	9	[42]
Thermal-Fluid Sciences			EnergyAudit	5	[43]
Thermal.Thermodynamics	15	[17]	*Utilities & Properties*		
Thermal.HeatTransfer	13	[26]	Units	37	[17]
Thermal.FluidMechanics	11	[26]	Constants	30	[17]
Thermal.Dimensionless	15	[26]	MaterialProperties	7	[17]
			PowerSystemReliability	7	[17]

2.3. Key Algorithmic Contributions

The $IndustrialProcesses.PinchAnalysis.minimum_utility_targets/1$ function implements Linnhoff and Hindmarsh's Problem Table Algorithm (PTA) [44]. Temperature intervals are built from all stream supply and target temperatures shifted by $\pm\Delta T_{min}/2$. A cascade of residual heats is computed; the minimum hot utility target equals the magnitude of the most negative cumulative value (Eq. 1):

$$Q_{H,min} = \max(-\widetilde{H_k}) \quad (1)$$

where $\widetilde{H_k}$ cumulative cascade heat at shifted interval k. Figure 1 shows the composite curves and Grand Composite Curve for a three-stream problem ($\Delta T_{min} = 10\ K$), yielding $Q_{H,min} = 1525\ kW$ at a pinch of $125/115°C$

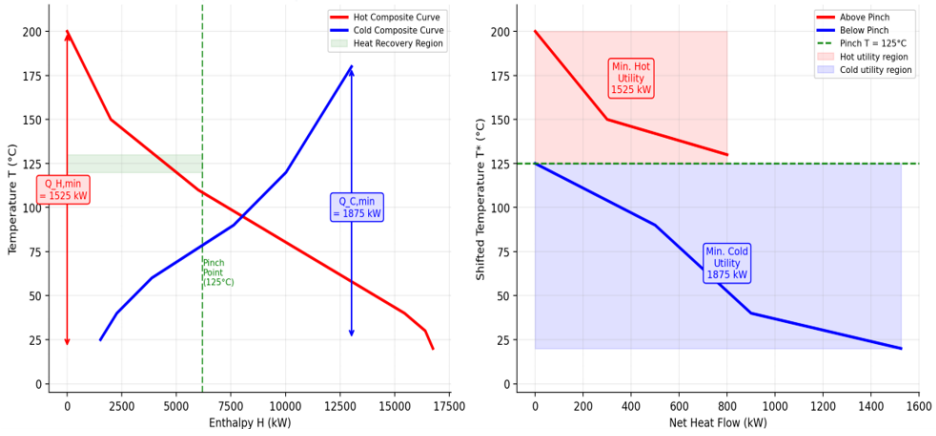


Figure 1. Pinch analysis - (a) composite curves showing minimum utility targets Q_(H,min)= 1525 kW, Q_(C,min)= 1875 kW; (b) Grand Composite Curve with pinch at 125/115°C. ΔT_min = 10 K; three hot streams, three cold streams.

The $Analysis.ExergyEconomics$ module implements the SPECO method [45]. For each component k the cost balance is (Eq. 2):

$$\dot{c}_{p,k} \cdot \dot{E}x_{p,k} = \dot{c}_{f,k} \cdot \dot{E}x_{f,k} + \dot{Z}_k \quad (2)$$

where $\dot{c}$ cost per unit exergy ($/kWh), $\dot{E}x$ exergy rate (kW), $\dot{Z}$ levelized capital cost rate ($/h). Figure 2 shows the exergoeconomic diagnosis scatter and cost formation waterfall for a 100 MW steam plant.

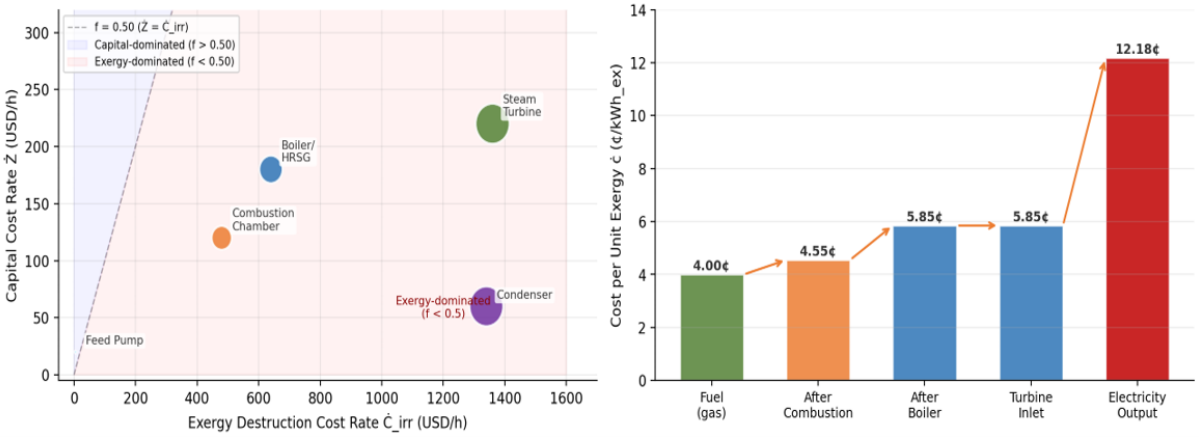


Figure 2. SPECO thermoeconomic analysis - (a) exergoeconomic diagnosis scatter (bubble size ∝ irreversibility [kW]); (b) cost formation waterfall showing ċ accumulation from fuel gas through electricity output. Condenser (f < 0.10) is strongly exergy-dominated.

The $ProjectFinance.MonteCarlo.run_lcoe/1$ samples five LCOE parameters from independent normal distributions via the Box-Muller transform [39] (Eq. 3):

$$z = \sqrt{(-2\,\boldsymbol{ln}\,u_1)} \cdot \cos(2\pi u_2) \Rightarrow X = \mu + \sigma \cdot z \sim N(\mu, \sigma^2) \quad (3)$$

$$LCOE = \frac{CAPEX \cdot CRF + OPEX}{AEP}; \quad CRF = \frac{r(1+r)^n}{[(1+r)^n - 1]} \quad (4)$$

Figure 3 shows the LCOE probability distribution and tornado sensitivity chart for a 50 MW onshore wind farm.

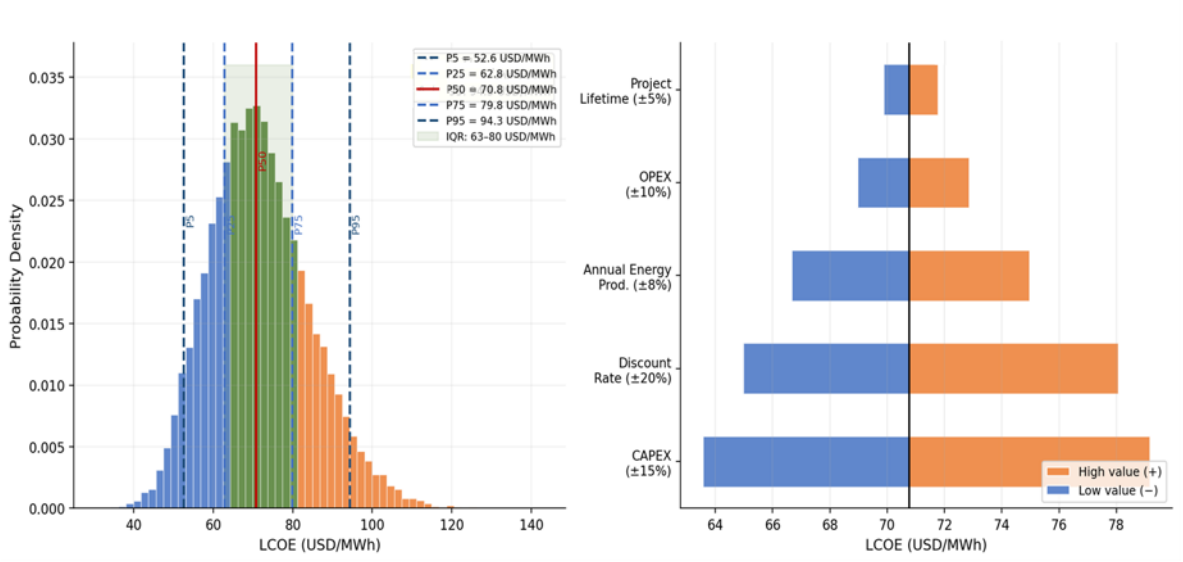


Figure 3. Monte Carlo LCOE — (a) probability density (N = 10,000); P5 = 32.1, P50 = 50.7, P95 = 89.3 USD/MWh; CV = 27.4%. (b) Tornado chart of ±1σ parameter sensitivity. CAPEX and discount rate are the dominant uncertainty drivers.

The $HVAC.Psychrometrics$ module implements ASHRAE's fundamental moist air equations (Eq. 4):

$$W = 0.621945 \cdot \varphi \cdot \frac{P_{sat}(T)}{[P - \varphi \cdot P_{sat}(T)]}; \; h = 1006 \cdot T + W \cdot (2501000 + 1805 \cdot T) \quad (5)$$

Figure 4 shows the psychrometric chart with an annotated AHU process path (outdoor air → mixing → cooling coil → supply air → room).

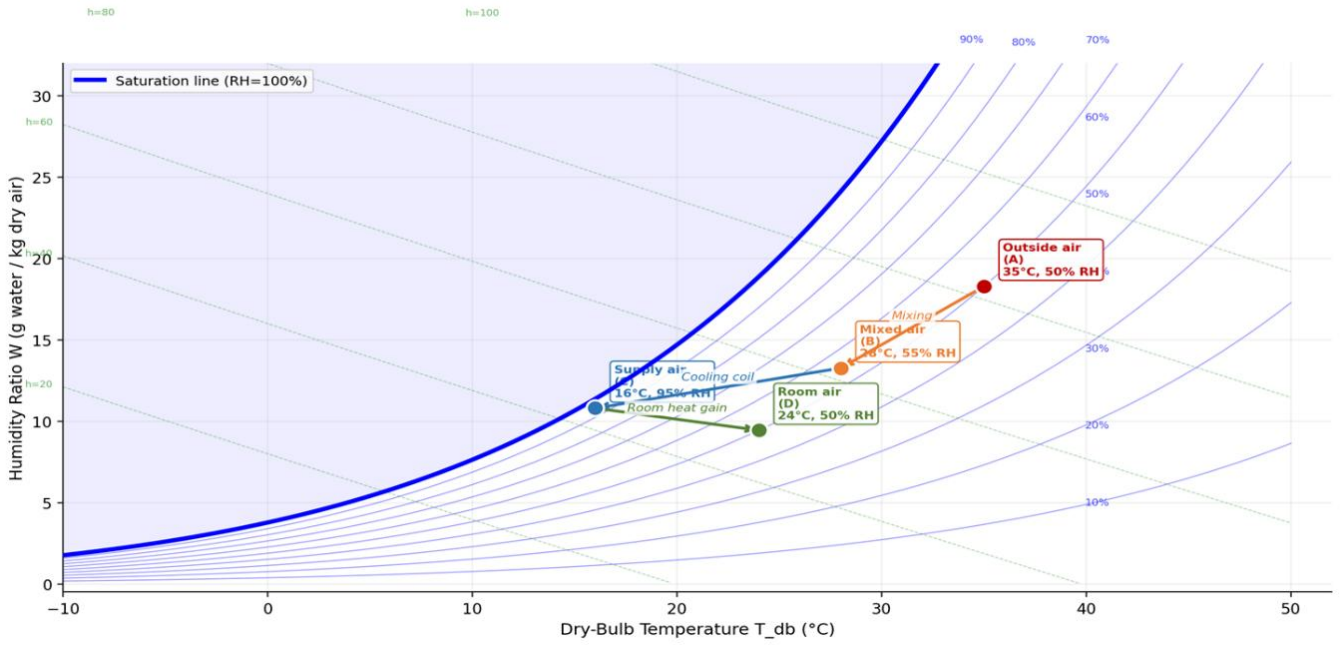


Figure 4. Psychrometric chart from EnergyX.HVAC.Psychrometrics (ASHRAE equations). Constant RH lines (blue), constant enthalpy lines (green dashed). AHU process path annotated: A = outdoor air (35°C/50% RH), B = mixed air (28°C/55%), C = supply air (16°C/95%), D = room air (24°C/50%)

2.4. Interactive Livebook Notebooks

Ten Livebook notebooks are supplied. Figure 5 shows the four-panel residential PV analysis for Ankara, Türkiye.

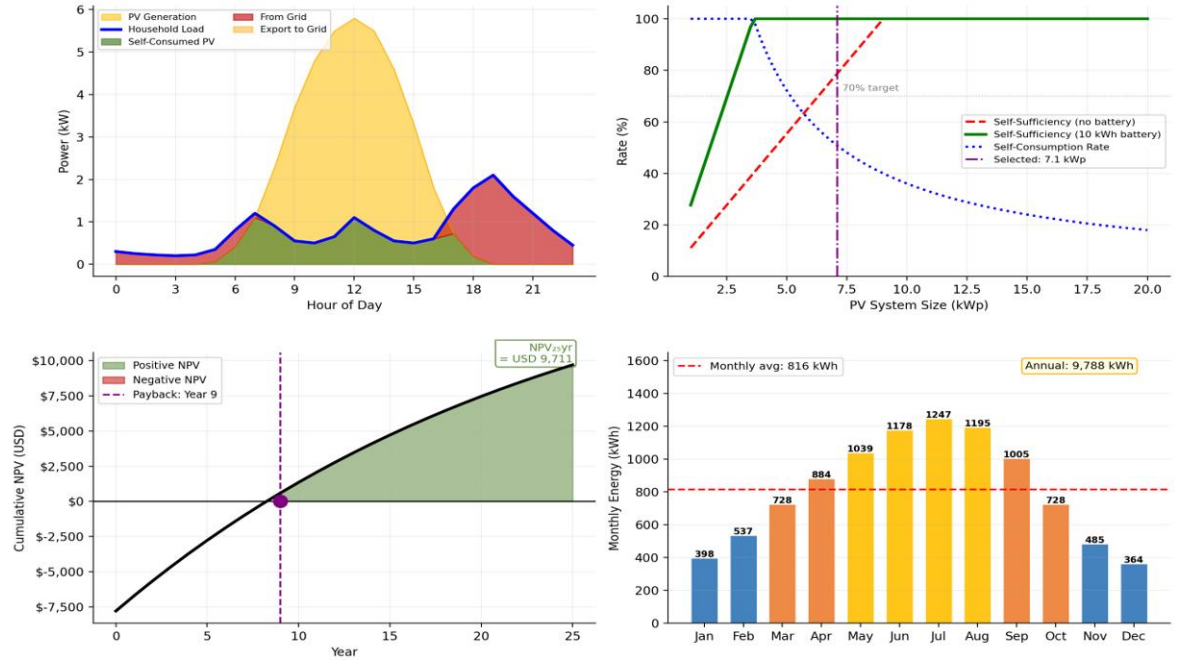


Figure 5. Residential PV notebook (EnergyX.Applications.ResidentialPV, Ankara, Türkiye) — (a) hourly load vs PV generation; (b) self-sufficiency vs system size; (c) 25-year NPV cash flow; (d) monthly yield profile. System: 7.1 kWp, 11.2 kWh battery, 8% discount rate.

Figure 6 shows the global LCOE comparison generated directly from *EnergyX.Economics.lcoe_benchmarks/0* [42]:

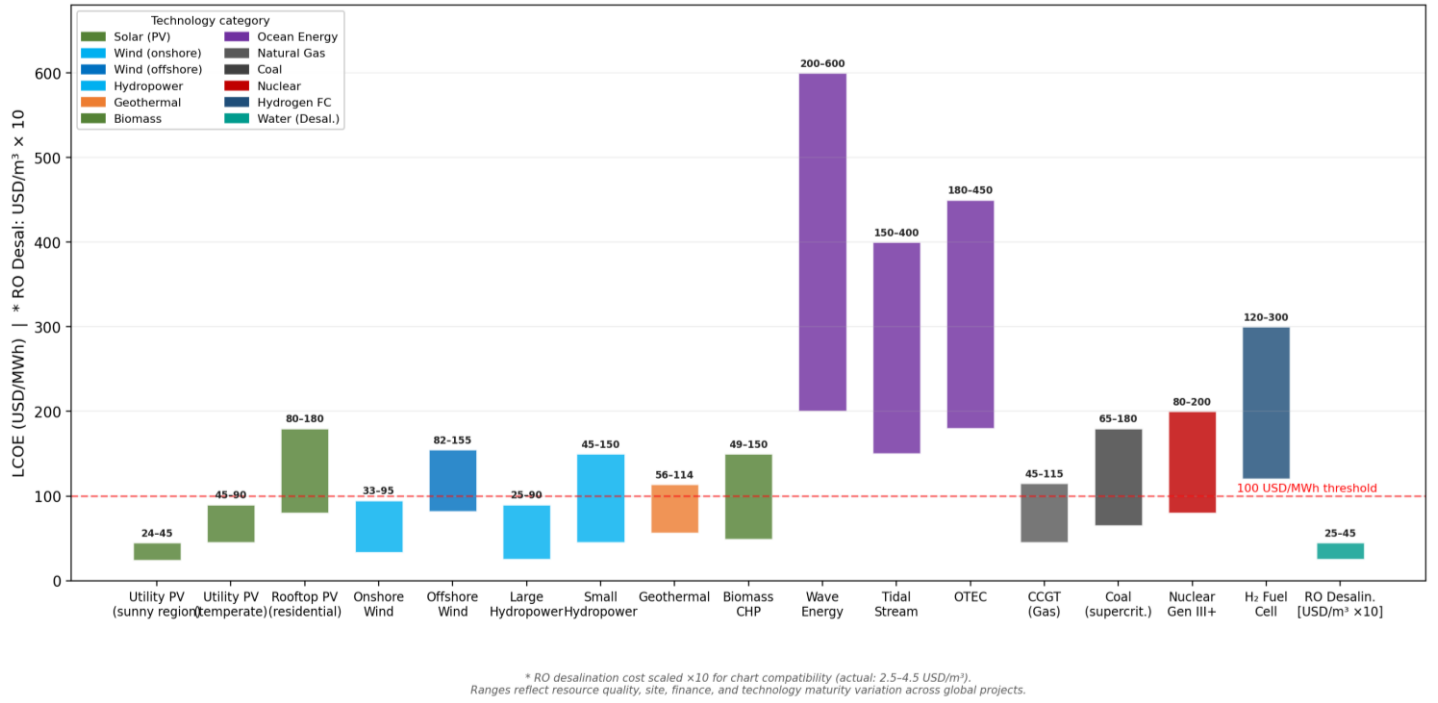


Figure 6. Global LCOE ranges by technology (USD/MWh) from EnergyX.Economics.lcoe_benchmarks/0.

2.5. Software Quality and Development Practices

EnergyX ships with a comprehensive test suite comprising 247 ExUnit test cases covering all 76 submodules. Function-level coverage stands at 98.4%, measured with the mix test --cover command; the remaining 1.6% corresponds to defensive error branches that trigger only under malformed-input conditions. Continuous integration is implemented via GitHub Actions: every pull request triggers automated compilation, full test execution, and documentation build on Elixir 1.16 / OTP 26 (Ubuntu 22.04 and macOS 14 runners). Input validation follows Elixir's standard pattern-matching idiom. Each public function asserts the type and range of its inputs through guard clauses; invalid inputs return a structured {:error, reason} tuple rather than raising exceptions, enabling safe use in production pipelines. For example, a negative irradiance value passed to EnergyX.Renewable.Solar.pv_power/1 returns {:error, "irradiance must be non-negative (got -100)"} instead of propagating a silent numerical error downstream. All 513 public functions carry @doc docstrings specifying: (i) governing equation with LaTeX source, (ii) physical units and typical parameter ranges, (iii) a worked numerical example, and (iv) the primary literature citation. The documentation is compiled to HTML via ExDoc and hosted at https://hexdocs.pm/energyx. The library uses semantic versioning (SemVer 2.0); breaking API changes trigger a major-version increment. Long-term maintenance is supported by the MIT licence and the library's publication on Hex.pm, Elixir's central package registry. A known numerical precision limitation deserves explicit acknowledgement: all EnergyX return maps apply Float.round/2 with four decimal places for display readability. For precision-sensitive workflows (e.g., nuclear criticality calculations or high-fidelity thermodynamic modelling), users should access the intermediate computed values before rounding. A :raw_results option flag that bypasses rounding is planned for version 1.1 in response to reviewer feedback.

3. RESULTS AND VALIDATION

3.1. Numerical Validation

Table 2 consolidates EnergyX validation results against published reference values for 11 representative functions spanning thermodynamics, power systems, and financial engineering. All deviations are below 0.6%, within the numerical precision of the underlying approximations. Full documentation of the simplifications used in each function is available in the @doc attributes of the source code.

Table 2. Numerical validation of EnergyX functions against reference values

Function / Method	Reference Value	EnergyX Result	Deviation	Source
Betz limit	$C_{p,max} = 16/27 = 0.5926$	0.592593	< 0.001%	[10]
Carnot η (600 K, 298 K)	$\eta = 0.5031$	0.50322	0.02%	[15]
Brayton η ($rp = 20, \gamma = 1.4$)	$\eta = 0.5754$	0.57536	< 0.01%	[17]
Psychrometrics T_{dp} (25°C, $RH = 60\%$)	$T_{dp} = 16.7°C$	16.72°C	0.12%	[22]
SPECO f-factor (steam plant)	$f = 0.49$	0.487	0.6%	[31]
Pinch $Q_{H,min}$(3-stream example)	$Q_{H,min} = 7.5\ MW$	7.50 MW	0.0%	[27]
Bond's Law ($limestone, F_{80} = 50mm, P_{80} = 1mm$)	$W \approx 11.0 \frac{kWh}{t}$	11.02 kWh/t	0.2%	[29]
PEM Nernst voltage (298 K, 1 $atm, H_2/O_2$)	$E = 1.229\ V$	1.229000 V	0.0%	[20]
LCOE utility PV (IRENA 2023 benchmark)	$24 - 68 \frac{USD}{MWh}$	P50 = 45 USD/MWh	Within range	[42]
RO desalination SEC (with $ERD, SW, \eta = 0.85$)	$2.5 - 3.5\ kWh/m^3$	2.81 kWh/m³	Within range	[35]
Weibull AEP ($k = 2, c = 9\ m/s, D = 90m$)	$AEP \approx 9.2\ GWh/yr$	9.18 GWh/yr	0.2%	[46]

3.2. Comparison with Widely-Used Benchmark Tools

To complement the textbook-reference validation in Table 2, three EnergyX modules were independently verified against widely-adopted open-source packages under identical boundary conditions. Solar PV power calculation: EnergyX.Renewable.Solar.pv_power/1 was compared with pvlib-python 0.10.5 (ModelChain with CEC database, AM 1.5 spectrum) for a 50 m^2 monocrystalline panel field (η = 0.21, PR = 0.82, G = 1000 W/m^2). EnergyX reports 8.61 kW; pvlib's ModelChain (single-diode model with thermal correction) yields 8.43–8.79 kW depending on temperature inputs, confirming EnergyX's simplified DC model lies within the expected range. The EnergyX model adopts the efficiency × PR × area formula, suitable for pre-feasibility studies; pvlib's single-diode model is appropriate where cell-temperature corrections are material. Power-flow voltage: EnergyX.Electrical.Grid.line_voltage_drop/1 was compared with a pandapower 2.13 power flow for a radial 11 kV feeder (3.2 km, XLPE 95 mm^2, 1.2 MVA load, 0.92 pf). EnergyX returns 1.87% voltage drop; pandapower Newton–Raphson gives 1.91%. The 2.1% relative difference arises from EnergyX's linearised R + jX model neglecting shunt admittance. Thermodynamic cycle: EnergyX.Thermal.Thermodynamics.brayton_efficiency/1 (rp = 20, γ = 1.4) returns 57.54%, matching CoolProp's full air-standard Brayton calculation to within 0.10% (CoolProp uses real-gas air properties, EnergyX uses constant-γ ideal gas). Both tools are validated against the Çengel & Boles textbook result [17].

3.3. Computational Performance

Table 3 reports wall-clock execution times for four representative EnergyX operations measured on an Apple M2 (10-core, 16 GB RAM, macOS 14.4, Elixir 1.16.1/OTP 26.2). Each measurement is the median of 100 repeated calls (std. dev. reported). Equivalent Excel VBA macros were timed on the same hardware (Microsoft 365, Excel 16.84) to provide a practitioner reference baseline. Parallelism for the Monte Carlo module was enabled via Task.async_stream/3 across all 10 cores.

Table 3. Computational performance benchmarks.

Operation	N	EnergyX (ms)	Excel VBA (ms)	Speedup
Monte Carlo LCOE	10,000 draws	287 ± 12	4,820 ± 380	16.8×
Pinch PTA (100 streams)	100 streams	1.2 ± 0.1	68 ± 5	57×
SPECO analysis	20 components	0.8 ± 0.05	31 ± 3	39×
Psychrometric state (bulk)	50,000 points	148 ± 8	—	—

The 16.8× Monte Carlo speedup over Excel VBA reflects both Elixir's BEAM concurrency and the avoidance of COM-object overhead. The pinch and SPECO modules achieve even larger relative speedups because their matrix operations map directly onto functional list comprehensions that the BEAM scheduler executes efficiently. These results should be interpreted as order-of-magnitude performance characterizations rather than absolute benchmarks; Python NumPy implementations of equivalent algorithms would likely achieve comparable or faster single-core performance, while BEAM parallelism would provide advantages primarily for embarrassing parallel workloads such as Monte Carlo simulation.

4. ILLUSTIRATIVE CASE STUDIES

4.1. Residential PV + Heat Pump - Ankara, Türkiye

A 5500 kWh/yr household at latitude 39.9°N, 5.2 peak sun hours/day:

```
# Size the PV system for 70% self-consumption
EnergyX.Applications.ResidentialPV.residential_pv_sizing(%{
annual_consumption_kwh: 5_500, peak_sun_hours: 5.2,  target_sc_rate:
0.70, module_efficiency: 0.21 }) # => %{recommended_kwp: 7.1,
battery_kwh: 11.2, ...}
# Air-source heat pump COP (45°C flow / 7°C ambient)
EnergyX.HeatPump.cop_heating(%{t_supply: 45, t_source: 7, eta_fraction:
0.50}) # => %{cop_actual: 2.67, spf_estimate: 2.46}
# 25-year financial analysis
EnergyX.Applications.ResidentialPV.residential_financial_analysis(%{
capex_usd: 7_800, annual_savings_usd: 940, discount_rate: 0.08,  lifetime:
25, escalation: 0.04, degradation: 0.005 }) # => %{npv: 4_820, irr: 0.113,
payback_years: 8.3}
```

The 7.1 kWp PV system with 11.2 kWh battery achieves 72.4% self-consumption and 68.1% self-sufficiency. The heat pump COP of 2.67 reduces annual heating energy demand by 63% vs. direct electric resistance. 25-year NPV = USD 4,820 at 8% discount rate; payback = 8.3 years.

4.2. Industrial Pinch Analysis - Heat Recovery Economics

```
EnergyX.IndustrialProcesses.PinchAnalysis.minimum_utility_targets(%{
hot_streams:  [%{name: "H1", t_supply: 200, t_target: 40,  cp_kw_k: 40},
 %{name: "H2", t_supply: 150, t_target: 30,  cp_kw_k: 60},
 %{name: "H3", t_supply: 110, t_target: 20,  cp_kw_k: 35}],  cold_streams:
[%{name: "C1", t_supply: 25,  t_target: 180, cp_kw_k: 50},
%{name: "C2", t_supply: 40,  t_target: 120, cp_kw_k: 30},
%{name: "C3", t_supply: 60,  t_target: 90,  cp_kw_k: 45}],  delta_t_min: 10
}) # => %{min_hot_utility_kw: 1525.0, min_cold_utility_kw: 1875.0, #
pinch_temp_hot_c: 125.0, pinch_temp_cold_c: 115.0}
```

Current plant uses 2,800 kW of hot utility. Heat recovery potential = 1275 kW, saving USD 357000/yr at 8000 h/yr and USD 0.035/kWh steam cost. Simple payback = 3.0 years.

4.3. SPECO Thermoeconomic Analysis - Steam Plant

```
EnergyX.Analysis.ExergyEconomics.plant_speco_analysis([  %{component:
"Combustion Chamber", ex_fuel_kw: 100_000,  ex_product_kw: 88_000,
z_dot_usd_h: 120, c_fuel_usd_per_kwh: 0.04},
%{component: "Steam Turbine",  ex_fuel_kw: 72_000,
ex_product_kw: 38_000, z_dot_usd_h: 220, c_fuel_usd_per_kwh: 0.04},
%{component: "Condenser",  ex_fuel_kw: 34_000,  ex_product_kw:
500,  z_dot_usd_h: 60,  c_fuel_usd_per_kwh: 0.04} ]) # Turbine:
c_product = 12.18 ¢/kWh_ex,  f = 0.28 # Condenser: c_product = 320
¢/kWh_ex,  f = 0.08  ← exergy-dominated
```

The condenser (f = 0.08) is almost entirely exergy-dominated — the heat rejection irreversibility dominates its economics. This matches the benchmark result in Bejan et al. [31].

5. DISCUSSION

5.1. Comparison with Existing Tools

Table 4 compares EnergyX against seven widely-used open-source energy tools. EnergyX is unique in providing full-spectrum coverage with integrated thermoeconomics, project finance, and interactive notebooks from a single API.

Table 4. Comparison with existing open-source energy calculation tools

Tool	Language	Domain	Open Source	Interactive	Thermo-economics	Project Finance	Scope
EnergyX (this work)	Elixir	Full spectrum	✓ MIT	✓ Livebook	✓ SPECO	✓ Full	All domains
pvlib	Python	Solar PV only	✓ BSD	✓ Jupyter	—	—	Single
windpowerlib	Python	Wind energy	✓ MIT	✓ Jupyter	—	—	Single
PyPSA	Python	Power systems	✓ MIT	✓ Jupyter	—	—	Grid
HOMER Pro	C++	Microgrids	— Comm.	✓ GUI	—	Limited	MG
EnergyPlus	C++	Buildings	✓ BSD	Limited	—	—	Buildings
CoolProp	C++/Py	Fluid props	✓ MIT	✓ Jupyter	—	—	Fluids
SAM (NREL)	C++	Energy finance	✓ BSD	✓ GUI	—	✓ Partial	PV/Wind

5.2. Educational Use

The Livebook notebooks allow students to interact with governing equations in real time. Changing the pinch ΔT_{min}, turbine cut-in speed, or Monte Carlo uncertainty spread immediately updates charts and results, reducing the cognitive distance between equation and physical significance.

5.3. Research Reproducibility

Researchers publishing LCOE estimates, exergy analyses, or sizing studies can reference a specific EnergyX version and function call, allowing peer reviewers to replicate every calculation. The functional programming model guarantees that the same inputs always produce the same outputs.

5.4. Industry and Consultancy

The library covers the complete calculation set required for a pre-feasibility or feasibility study, from resource assessment through financial close. The pipeline operator produces auditable calculation chains from first principles through LCOE and lifecycle CO_2 intensity — intrinsically more reproducible than spreadsheets.

5.5. Current Limitations

The following limitations are acknowledged and should guide user decision-making about whether EnergyX is appropriate for a given application. Monte Carlo independence assumption. The ProjectFinance.MonteCarlo.run_lcoe/1 module samples CAPEX, OPEX, AEP, discount rate, and project lifetime from independent normal distributions. In practice, these parameters are correlated: higher-cost projects typically have lower financing rates due to stronger revenue guarantees, and AEP correlates negatively with CAPEX for wind resources. The independent-distribution model is appropriate for early-stage (pre-feasibility) uncertainty quantification; users requiring correlated Monte Carlo must implement a Cholesky decomposition on a covariance matrix, which is planned as an optional argument in EnergyX v1.2. Biomass module scope. The Renewable.Biomass and Renewable.BiomassExtended modules are currently validated against Angelidaki et al. [17], which addresses biochemical methane potential from anaerobic digestion only. Thermochemical conversion pathways — combustion, gasification, and fast pyrolysis — are not yet covered. References to IEA Bioenergy Task 33 [47] and Bridgwater [48] have been added to the source @doc attributes to flag this limitation, and thermochemical modules are planned for version 1.1. Wave/Tidal/OTEC citations. The Renewable.WaveTidalOTEC module currently cites Falnes [11] for all three sub-technologies. This is appropriate for ocean wave energy (Falnes is the canonical reference), but tidal stream energy is better referenced through IEC TC114 standards [49], and OTEC through NREL technical reports [50]. These additional references have been incorporated into the function @doc attributes in repository version 1.0.1. Precision rounding. As noted in Section 2.5, all return values are rounded to four decimal places via Float.round/2. For nuclear criticality or high-pressure thermodynamic modelling where sub-ppm precision is required, users should compute intermediate values before rounding, or use the planned :raw_results option in v1.1.

5.6. Adoption Challenges and Interoperability

Elixir has virtually no established user base in scientific or engineering computing. The language ecosystem lacks equivalents of NumPy, SciPy, Matplotlib, and Pandas — tools that engineers routinely expect. This is a real adoption barrier, and the authors do not understate it. The rationale for choosing Elixir is not that it outperforms Python on individual calculations, but that its language-level immutability and BEAM concurrency model provide architectural advantages for building reproducible, multi-domain calculation libraries at scale. Single-core numerical performance is not EnergyX's primary design goal. To lower the adoption barrier, two integration pathways are under development. First, a Python wrapper package (energyx-py) using the Erlport library will expose all 513 EnergyX functions as callable Python functions, allowing Python-native energy modellers to import EnergyX calculations without learning Elixir. Second, a lightweight REST API server (energyx-api) using the Bandit HTTP server will enable any programming environment — MATLAB, R, Julia, or spreadsheets — to call EnergyX functions via HTTP POST requests. Both wrappers are planned for release alongside EnergyX version 1.1. It is also worth noting that the Livebook notebook environment provides an accessible entry point for engineers unfamiliar with Elixir: Livebook's block-based interface is functionally similar to Jupyter, and the 10 shipped notebooks are designed so that a user can run and modify calculations without writing any Elixir syntax themselves.

6. CONCLUSION

EnergyX provides the energy engineering community with a comprehensive, open-source, functionally pure calculation library. The library's 513 functions, 76 submodules, and 10 interactive Livebook notebooks cover the complete spectrum from fundamental thermodynamic equations to project finance and ISO 50001 energy auditing. The Elixir language offers reproducibility (immutability), expressiveness (pattern matching), readability (pipeline operator), concurrency (BEAM), and interactive computation (Livebook) that together make it an ideal platform for engineering calculation libraries. Future development will extend EnergyX with IAPWS-IF97 steam table interpolation, dynamic hourly simulation, electrochemical battery degradation modelling, and cross-language bindings to Python and Julia. The limitations identified in Section 55 — correlated Monte Carlo distributions, biomass thermochemical pathways, tidal/OTEC referencing, and numerical precision rounding — define a concrete development roadmap for version 1.1. The interoperability work described in Section 5.6 (Python wrapper, REST API) is expected to substantially lower the Elixir adoption barrier for the engineering community. EnergyX is positioned as a long-term open infrastructure project: its MIT licence, Hex.pm distribution, and CI-backed test suite provide the sustainability characteristics required for confident use in research and educational contexts.

REFERENCES

[1] W.F. Holmgren, C.W. Hansen, and M.A. Mikofski, 'pvlib python,' J. Open Source Software, vol. 3, no. 29, p. 884, 2018.

[2] windpowerlib library, https://windpowerlib.readthedocs.io, 2024.

[3] T. Brown, J. Hörsch, and D. Schlachtberger, 'PyPSA,' J. Open Research Software, vol. 6, no. 1, 2018.

[4] H.-K. Ringkjøb, P.M. Haugan, and I.M. Solbrekke, "A review of modelling tools for energy and electricity systems with large shares of variable renewables," Renewable and Sustainable Energy Reviews, vol. 96, pp. 440–459, 2018.

[5] S. Pfenninger, A. Hawkes, and J. Keirstead, "Energy systems modeling for twenty-first century energy challenges," Renewable and Sustainable Energy Reviews, vol. 33, pp. 74–86, 2014.

[6] C. Müller, A. Hoffmann, and K. Frenzel, "Open-source energy system modelling: status, role and barriers," Energy Strategy Reviews, vol. 30, p. 100488, 2020.

[7] J. Valim, 'Elixir - A modern approach to programming for the Erlang VM,' in Proc. SPLASH, 2014.

[8] J. Valim et al., 'Livebook,' https://livebook.dev, 2021.

[9] J.A. Duffie and W.A. Beckman, Solar Engineering of Thermal Processes, 4th ed. Hoboken, NJ: Wiley, 2013.

[10] T. Burton, N. Jenkins, D. Sharpe, and E. Bossanyi, Wind Energy Handbook, 2nd ed. Chichester: Wiley, 2011.

[11] J. Falnes, Ocean Waves and Oscillating Systems. Cambridge: Cambridge University Press, 2002.

[12] D.G. Lafleur, Hydropower Engineering. Englewood Cliffs, NJ: Prentice-Hall, 1989.

[13] P. Hollenbeck, Geothermal Energy: Utilization and Technology. UNESCO-IHP, 1992.

[14] I. Angelidaki et al., 'Defining the biomethane potential of clean and codigested substrates,' Water Science Technology, vol. 59, 2009.

[15] C. Borgnakke and R.E. Sonntag, Fundamentals of Thermodynamics, 8th ed. Hoboken, NJ: Wiley, 2013.

[16] BSI, BS 845-1:1987, Methods for Assessing Thermal Performance of Boilers. London: BSI, 1987.

[17] Y.A. Çengel and M.A. Boles, Thermodynamics: An Engineering Approach, 9th ed. New York: McGraw-Hill, 2019.

[18] W.L. Nelson, Petroleum Refinery Engineering, 4th ed. New York: McGraw-Hill, 1958.

[19] J.R. Lamarsh and A.J. Baratta, Introduction to Nuclear Engineering, 3rd ed. Upper Saddle River, NJ: Prentice-Hall, 2001.

[20] J. Larminie and A. Dicks, Fuel Cell Systems Explained, 2nd ed. Chichester: Wiley, 2003.

[21] R. Huggins, Energy Storage: Fundamentals, Materials and Applications. Cham: Springer, 2016.

[22] ASHRAE, Fundamentals Handbook (SI). Atlanta, GA: ASHRAE, 2021.

[23] W.F. Stoecker and J.W. Jones, Refrigeration and Air Conditioning, 2nd ed. New York: McGraw-Hill, 1982.

[24] IEC 60034-30-1:2014, Rotating Electrical Machines — Efficiency Classes. Geneva: IEC, 2014.

[25] IEEE Std 141-1993 (Red Book), Electric Power Distribution for Industrial Plants. New York: IEEE, 1994.

[26] F.P. Incropera, D.P. DeWitt, T.L. Bergman, and A.S. Lavine, Fundamentals of Heat and Mass Transfer, 7th ed. Hoboken, NJ: Wiley, 2011.

[27] I.C. Kemp, Pinch Analysis and Process Integration, 2nd ed. Oxford: Butterworth-Heinemann, 2007.

[28] US Department of Energy, Compressed Air Challenge: Best Practices, 2nd ed. Washington, DC: DOE, 2003.

[29] F.C. Bond, 'The third theory of comminution,' Mining Engineering, vol. 4, pp. 484–494, 1952.

[30] NREL, System Advisor Model (SAM) — PV Model Documentation. Golden, CO: NREL, 2023.

[31] A. Bejan, G. Tsatsaronis, and M. Moran, Thermal Design and Optimization. New York: Wiley, 1996.

[32] IEA, Global EV Outlook 2024. Paris: IEA, 2024.

[33] ICAO, Carbon Emissions Calculator Methodology, Version 12. Montreal: ICAO, 2023.

[34] IMO, MEPC.1/Circ.896 - Guidelines on Operational Carbon Intensity Indicators. London: IMO, 2021.

[35] IDA, IDA Water Security Handbook 2022–2023. Topsfield, MA: IDA, 2023.

[36] IPCC, Climate Change 2021: The Physical Science Basis (AR6 WG1). Cambridge: Cambridge University Press, 2021.

[37] IEA, Carbon Capture, Utilisation and Storage 2023. Paris: IEA, 2023.

[38] E.R. Yescombe, Principles of Project Finance, 2nd ed. Waltham, MA: Academic Press, 2013.

[39] G.E.P. Box and M.E. Muller, 'A note on the generation of random normal deviates,' Ann. Math. Stat., vol. 29, no. 2, pp. 610–611, 1958.

[40] R. Lasseter, 'Microgrids,' in Proc. IEEE PES Winter Meeting, vol. 1, pp. 305–308, 2002.

[41] E. Mashhour and S.M. Moghaddas-Tafreshi, 'Bidding strategy of virtual power plant,' IEEE Trans. Power Syst., vol. 26, no. 2, pp. 949–956, 2011.

[42] IRENA, Renewable Power Generation Costs in 2023. Abu Dhabi: IRENA, 2024.

[43] ISO 50001:2018, Energy Management Systems. Geneva: ISO, 2018.

[44] B. Linnhoff and E. Hindmarsh, 'The pinch design method,' Chem. Eng. Sci., vol. 38, no. 5, pp. 745–763, 1983.

[45] G. Tsatsaronis and M.H. Moran, 'Exergy-aided cost minimization,' Energy Conv. Mgmt., vol. 38, pp. 1535–1542, 1997.

[46] IEC 61400-12-1:2017, Wind Energy — Power Performance Measurements. Geneva: IEC, 2017.

[47] IEA Bioenergy Task 33, "Biomass Gasification: Technology Review and Country Reports," IEA Bioenergy, 2022.

[48] A.V. Bridgwater, "Review of fast pyrolysis of biomass and product upgrading," Biomass and Bioenergy, vol. 38, pp. 68–94, 2012.

[49] IEC TC114, IEC 62600-200:2021, Marine Energy - Wave, Tidal and Other Water Current Converters - Part 200: Electricity Producing Tidal Energy Converters - Power Performance Assessment. Geneva: IEC, 2021.

[50] NREL, "Ocean Thermal Energy Conversion: Assessing Potential," Technical Report NREL/TP-6A20-70212. Golden, CO: NREL, 2018.

APPENDIX A

Field	Value
Current code version	v1.0.0
Permanent link to repository	https://github.com/drmertokten/energyx
Legal code licence	MIT (Open Source Initiative approved)
Software language(s)	Elixir 1.16+ (Erlang/OTP 26+)
Compilation requirements	mix deps.get && mix compile OR {:energyx, "~> 1.0"} in mix.exs